Making Models Unmergeable via Scaling-Sensitive Loss Landscape

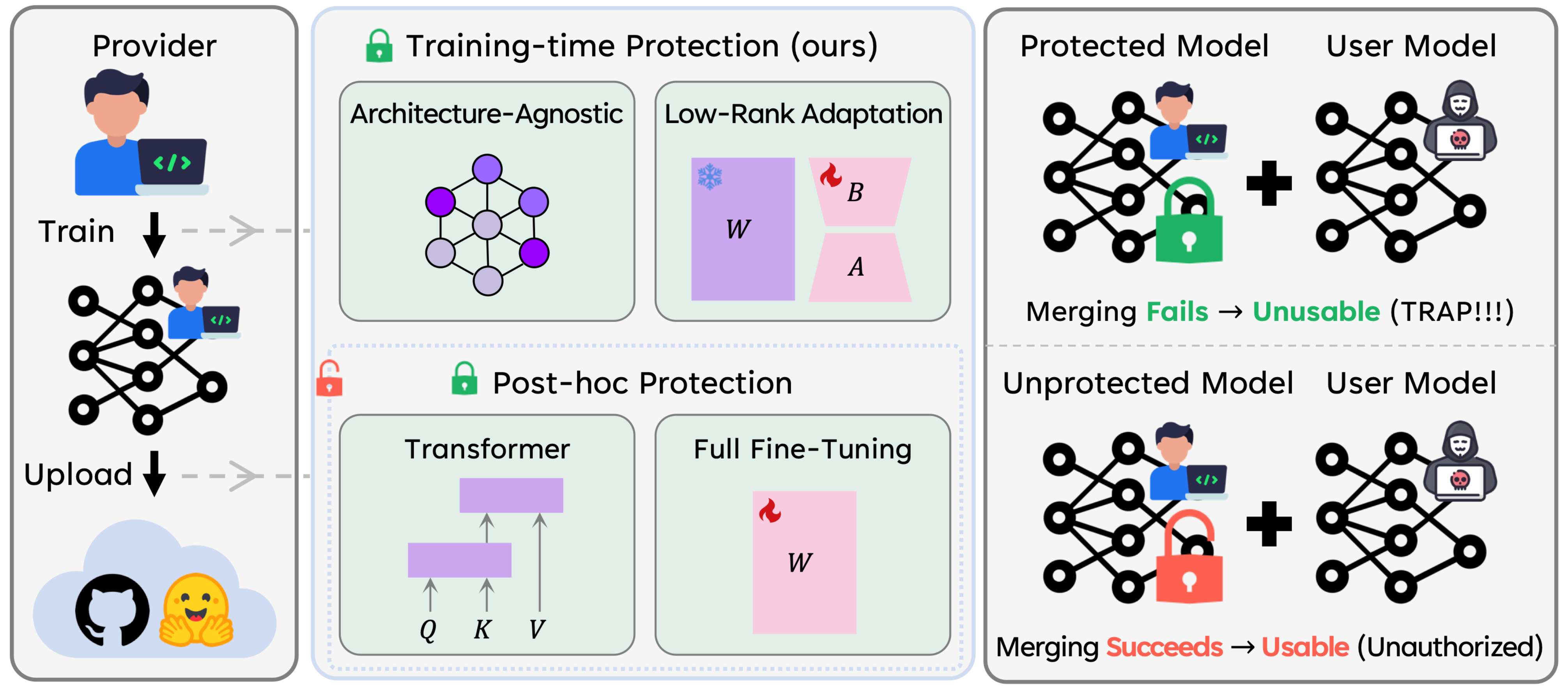
Minwoo Jang[P], Hoyoung Kim[N], Jabin Koo[P], Jungseul Ok[P]

[P] Pohang University of Science and Technology (POSTECH)

[N] National AI Research Lab (NAIRL)

Why do we need unmergeability?

Once released, model weights move **beyond your control**. When merged into composite models, their **provenance** is obscured, **license compliance** becomes unenforceable, and **safety alignment** is weakened.



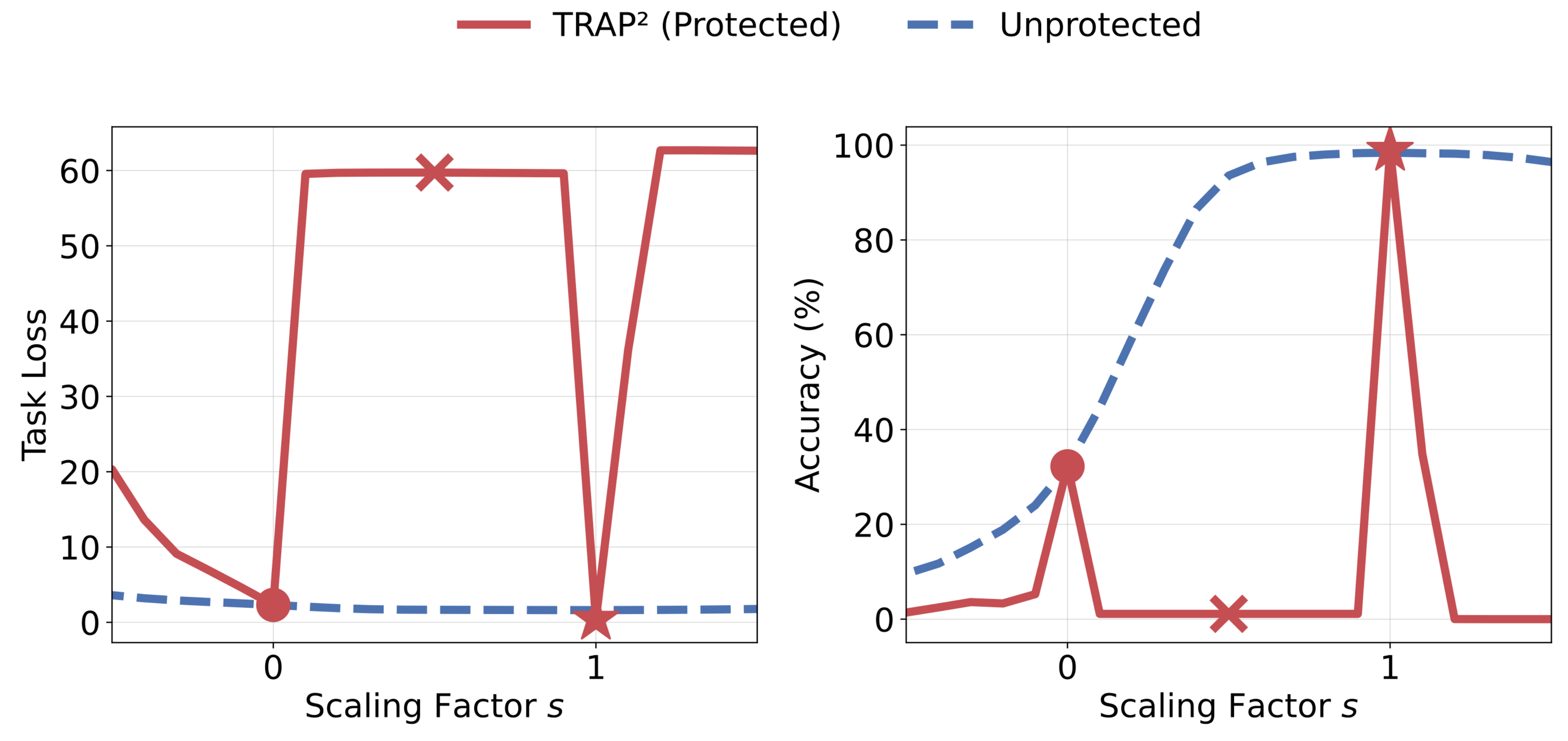
Why do we need Trap²?

(**Tra**ining-time **P**rotection via **T**ask-**R**obust **A**dversarial **P**erturbation)

Existing protections (e.g., Merge-Lock [1], PaRaMS [2]) assume **Transformer models with full-weight access**. However, many real-world releases can be **adapter-only** or **non-Transformer**. **Trap²** closes that gap.

Intuition: Update scaling as a proxy for merging

At the authorized scale ($s = 1$), the update remains stable and preserves standalone utility. However, when merged, **each update is often down-scaled**. For example, uniform averaging assigns each of N adapters a scale of $s = 1/N < 1$. Once **shifted away from the authorized scale** ($s \neq 1$), the update becomes **fragile**.



Trap² objective for LoRA $\Delta W = BA$

$$J(\Delta W) = \mathcal{L}_{nominal}(\Delta W) - \lambda \cdot \mathcal{L}_{off}(\Delta W)$$

$$\mathcal{L}_{nominal}(\Delta W) := \mathcal{L}(W_0 + \Delta W)$$

$$\mathcal{L}_{off}(\Delta W) := \mathbb{E}_{s \in [s_{min},\, 1-\delta] \cup [1+\delta,\, s_{max}]} \left[\frac{1}{s} \cdot \mathcal{L}(W_0 + s \cdot \Delta W) \right]$$

Extension of Trap² to Full Fine-Tuning

Just replace ΔW with $\Delta W_t := W_t - W_0$ at each step t.

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Standalone accuracy before merging (%; ↑)

Backbone	Method	Cars	DTD	EuroSAT	GTSRB	MNIST	RESISC	Aircraft	SVHN	Average
ViT-B/32	Zero-Shot	59.397	44.096	45.481	32.255	47.988	60.302	18.962	31.384	42.483
	Fine-Tuned	99.523	68.723	98.259	98.416	99.087	93.048	51.725	96.168	88.119
	+ Merge-Lock*	0.575	2.394	10.963	2.009	9.462	3.540	11.423	9.666	6.254
	+ Merge-Lock†	0.482	2.074	6.333	2.059	9.988	2.111	1.140	9.066	4.157
	+ PaRaMS*	97.839	67.713	98.296	98.238	99.125	92.476	51.305	96.111	87.638
	+ PaRaMS†	92.150	61.596	92.222	97.694	98.125	91.127	46.685	95.534	84.392
	TRAP² (Ours)	99.829	66.117	98.519	98.911	99.462	93.302	52.355	96.557	88.132
ViT-L/14	Zero-Shot	77.864	55.532	62.259	50.713	76.162	71.365	32.583	58.552	60.629
	Fine-Tuned	99.767	76.702	98.778	98.466	99.587	95.587	72.577	97.758	92.403
	+ Merge-Lock*	0.497	2.340	8.593	3.256	11.737	1.571	0.930	6.473	4.425
	+ Merge-Lock†	0.389	1.862	9.111	1.880	11.950	1.635	1.110	8.259	4.525
	+ PaRaMS*	99.518	76.170	98.593	98.357	99.550	94.698	69.967	43.815	85.084
	+ PaRaMS†	98.912	74.043	98.407	98.248	99.638	93.397	67.027	41.424	83.887
	TRAP² (Ours)	99.876	78.830	98.481	99.584	99.438	96.175	80.138	97.167	93.711
ConvNeXt	Zero-Shot	89.554	59.628	54.519	48.892	54.888	67.143	27.512	34.327	54.558
	Fine-Tuned	98.492	76.436	98.926	99.268	99.325	96.063	59.226	97.105	90.605
	TRAP² (Ours)	97.606	76.064	98.630	98.931	99.488	94.079	64.326	96.159	90.660

Averaged accuracy after merging (%; ↓)

Protection	Merging Method / Merging Space												
	TA	TIES			TIES+DARE			TSV			CART		
	Full	Full	KnOTS	Core	Full	KnOTS	Core	Full	KnOTS	Core	Full	KnOTS	Core
Backbone: ViT-B/32 (Zero-shot: 42.483)													
Unprotected	48.273	48.020	49.925	53.264	48.180	49.926	54.745	51.442	49.202	55.014	49.553	49.850	51.201
PaRaMS*	48.290	48.027	49.847	53.054	48.329	49.801	54.506	51.901	49.249	55.049	49.663	49.908	51.922
PaRaMS†	48.104	48.078	49.872	53.022	48.208	49.872	54.080	51.418	49.033	54.509	49.418	49.791	50.647
TRAP² (Ours)	23.121	36.480	37.514	37.213	33.880	28.383	36.752	24.815	24.210	24.449	41.324	40.967	41.602
Backbone: ViT-L/14 (Zero-shot: 60.629)													
Unprotected	62.914	67.909	68.879	68.825	67.970	69.870	68.779	72.411	66.725	74.565	64.399	65.014	66.185
PaRaMS*	62.926	67.821	69.583	69.229	67.915	69.631	69.215	71.985	66.740	74.592	64.358	65.012	66.231
PaRaMS†	62.826	67.875	69.581	69.287	67.877	69.596	69.302	72.150	66.355	74.097	64.262	64.811	65.825
TRAP² (Ours)	33.619	53.162	47.749	54.462	50.630	43.269	54.847	41.993	37.369	39.389	58.634	58.444	59.514
Backbone: ConvNeXt (Zero-shot: 54.558)													
Unprotected	49.203	59.603	60.201	63.601	59.620	60.305	63.573	65.133	60.351	65.836	60.069	59.997	61.215
TRAP² (Ours)	14.719	32.883	15.187	17.188	15.636	13.054	25.335	14.085	15.850	15.296	46.973	45.885	47.204

References:

[1] Model Unmerging: Making Your Models Unmergeable for Secure Model Sharing (arXiv 2025; Wang et al.)

[2] Disrupting Model Merging: A Parameter-Level Defense Without Sacrificing Accuracy (ICCV 2025; Junhao et al.)

Protection	Data-dependent Merging		Data-driven Recovery	
	RegMean	CoM	ProDistill	SFT
Backbone: ViT-B/32 (Zero-shot: 42.483)				
Unprotected	49.107	64.776	73.823	56.859
TRAP² (Ours)	9.480	32.331	49.153	45.496

Key Takeaways

Trap² extends unmergeability to **LoRA adapters** and **non-Transformer backbones**. It makes updates **scaling-sensitive**: useful alone, **fragile once merged**.